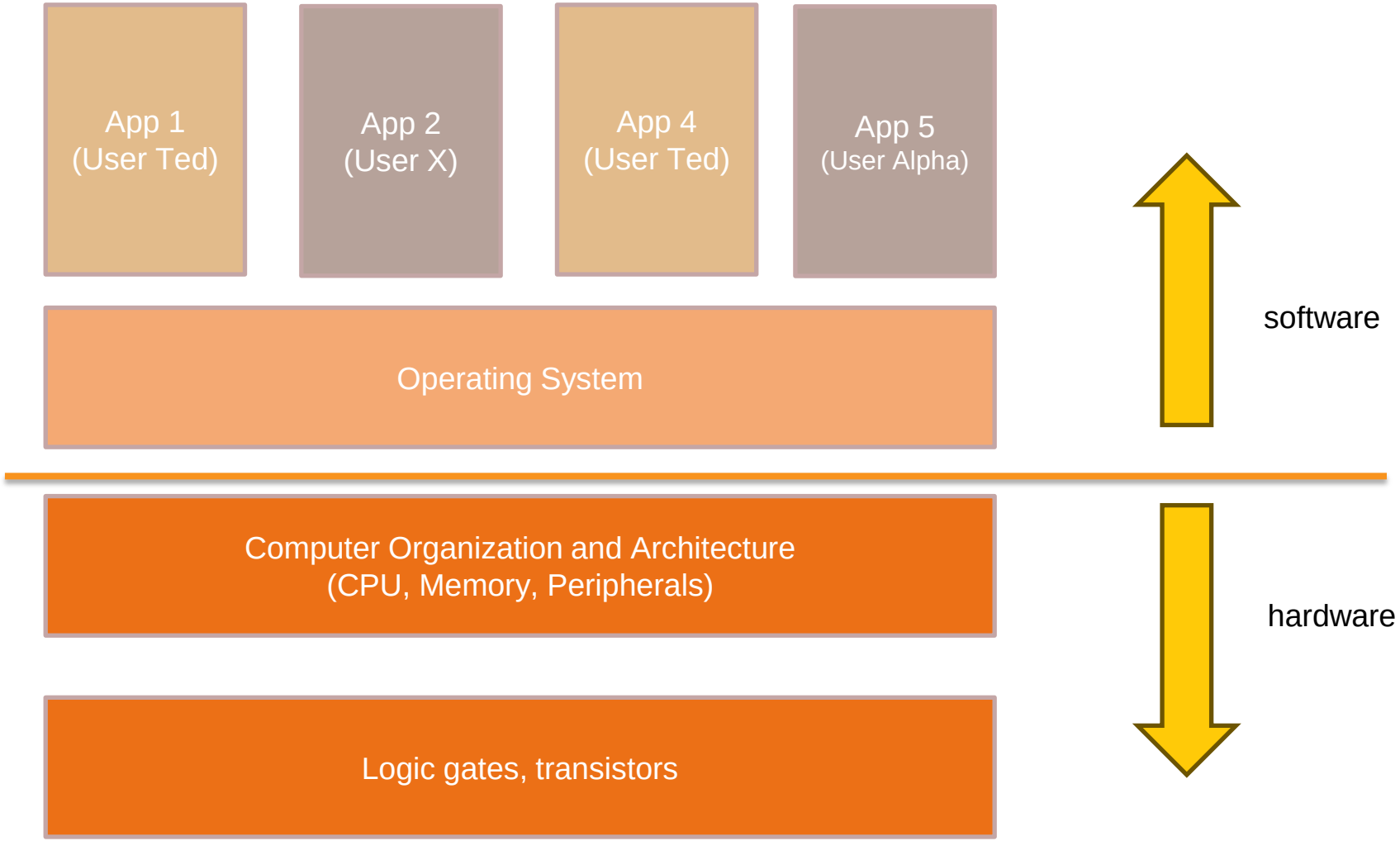
Computer Organization and Architecture (CSL251)

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The original slides taken from

Chester Rebeiro
Indian Institute of Technology Madras
chester@cse.iitm.ac.in

The Compute Stack



The Compute Stack

App 1
(User Ted)

App 2
(User X)

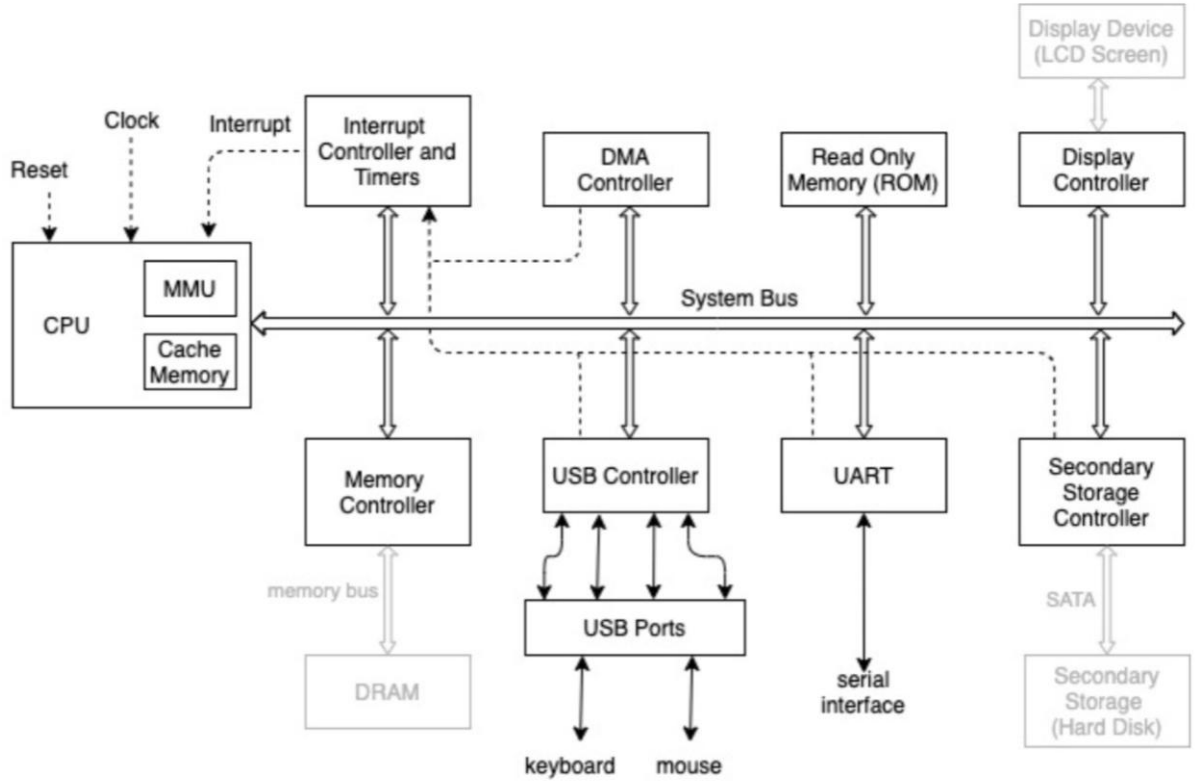
App 4
(User Ted)

App 5
(User Alpha)

Operating System

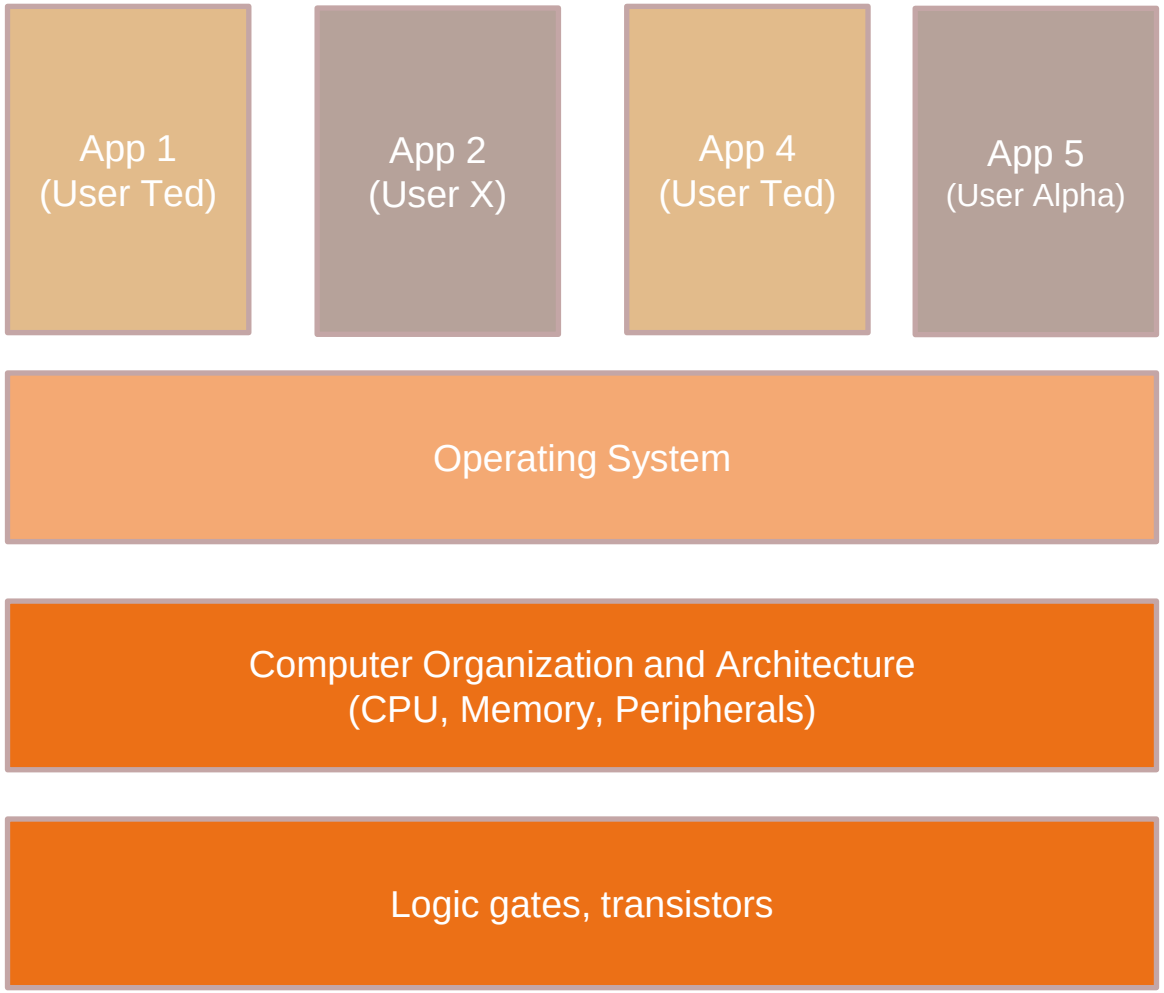
Computer Organization and Architecture
(CPU, Memory, Peripherals)

Logic gates, transistors

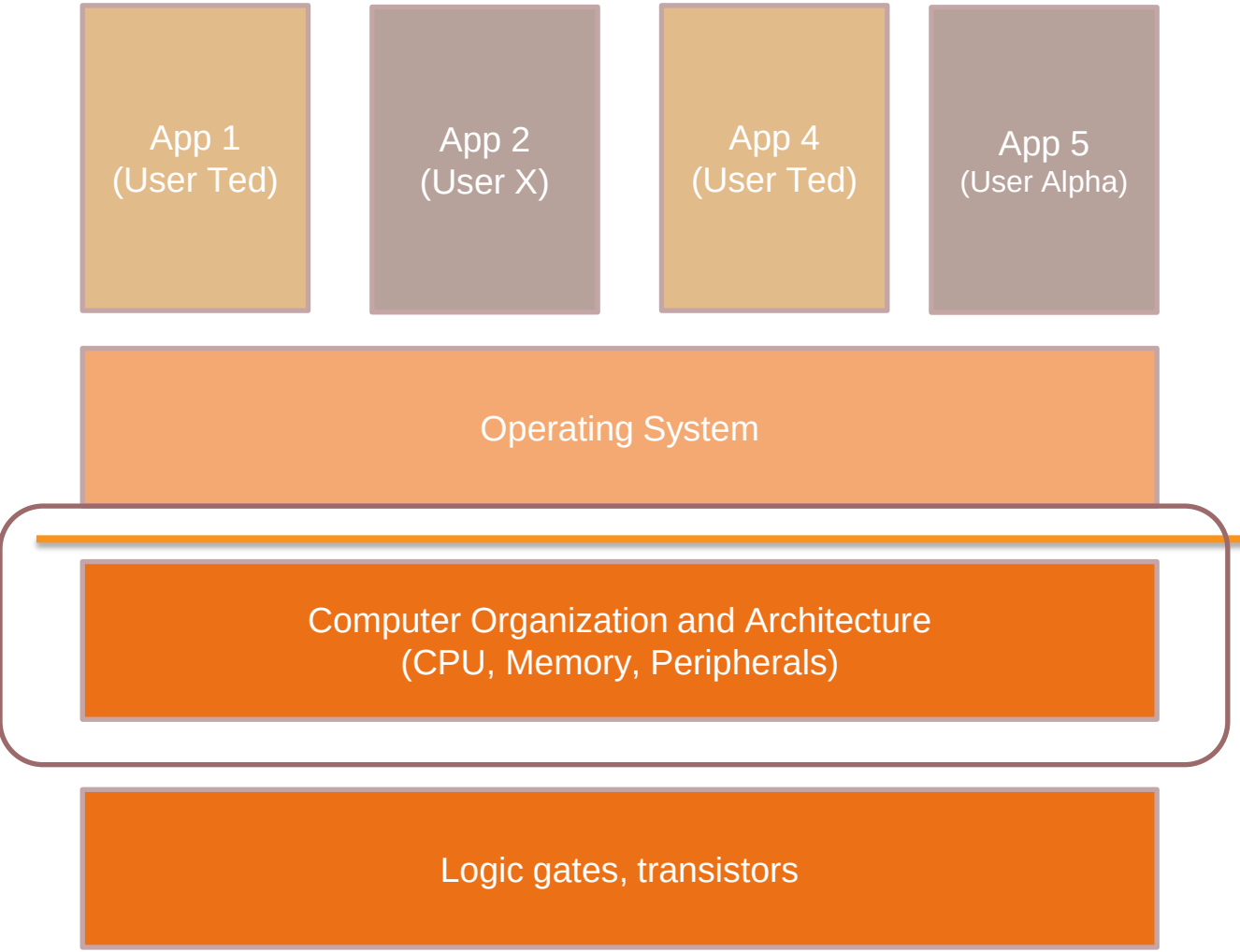


The Wish List

- Fast
- Energy Efficient
- Program support
- Security and Protection
- Multi-user systems

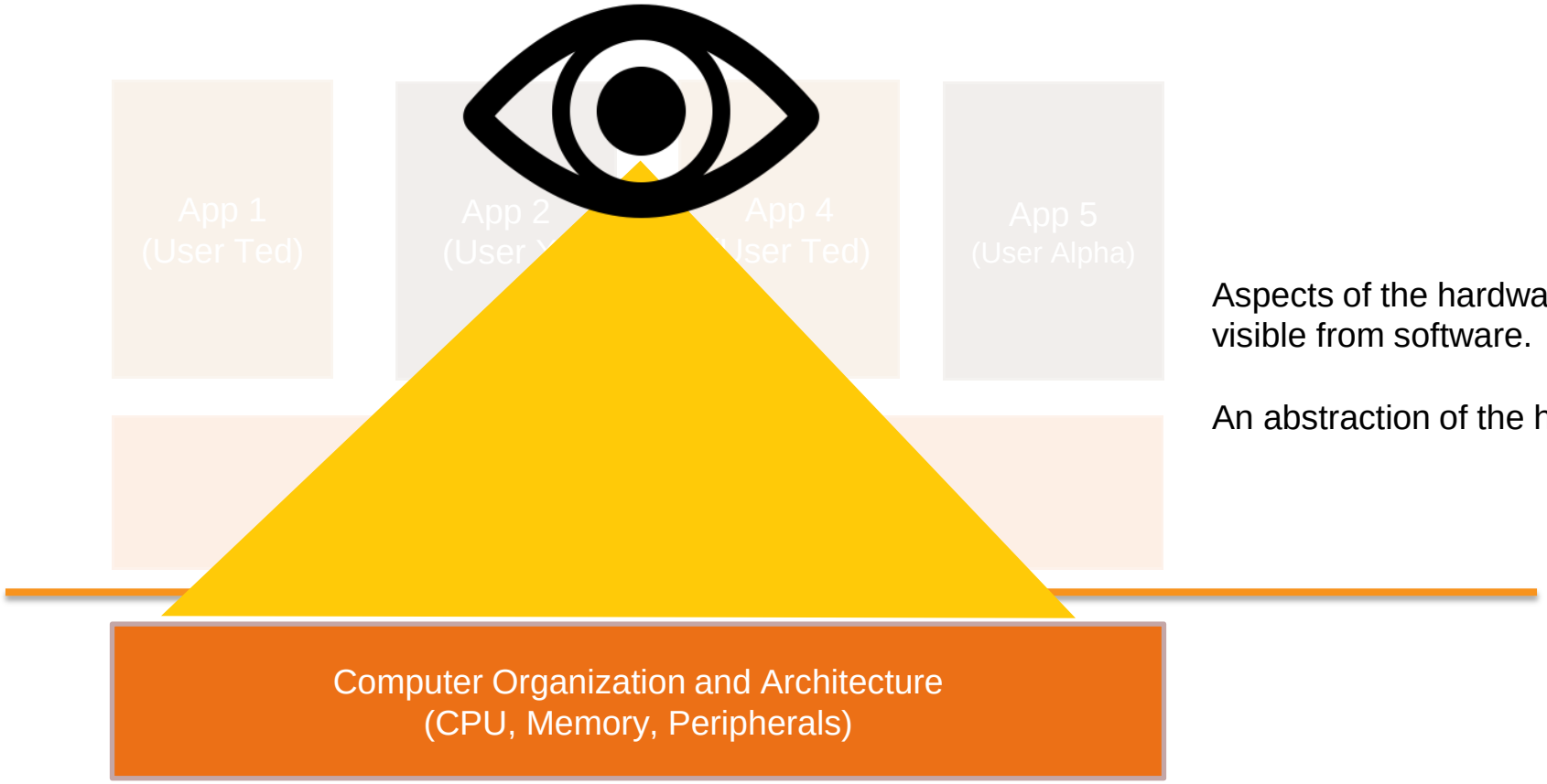


Where does COA fit in the Compute Stack



The layer which converts the logic gates into modules that can be usable by software to achieve the wishlist

Computer Architecture

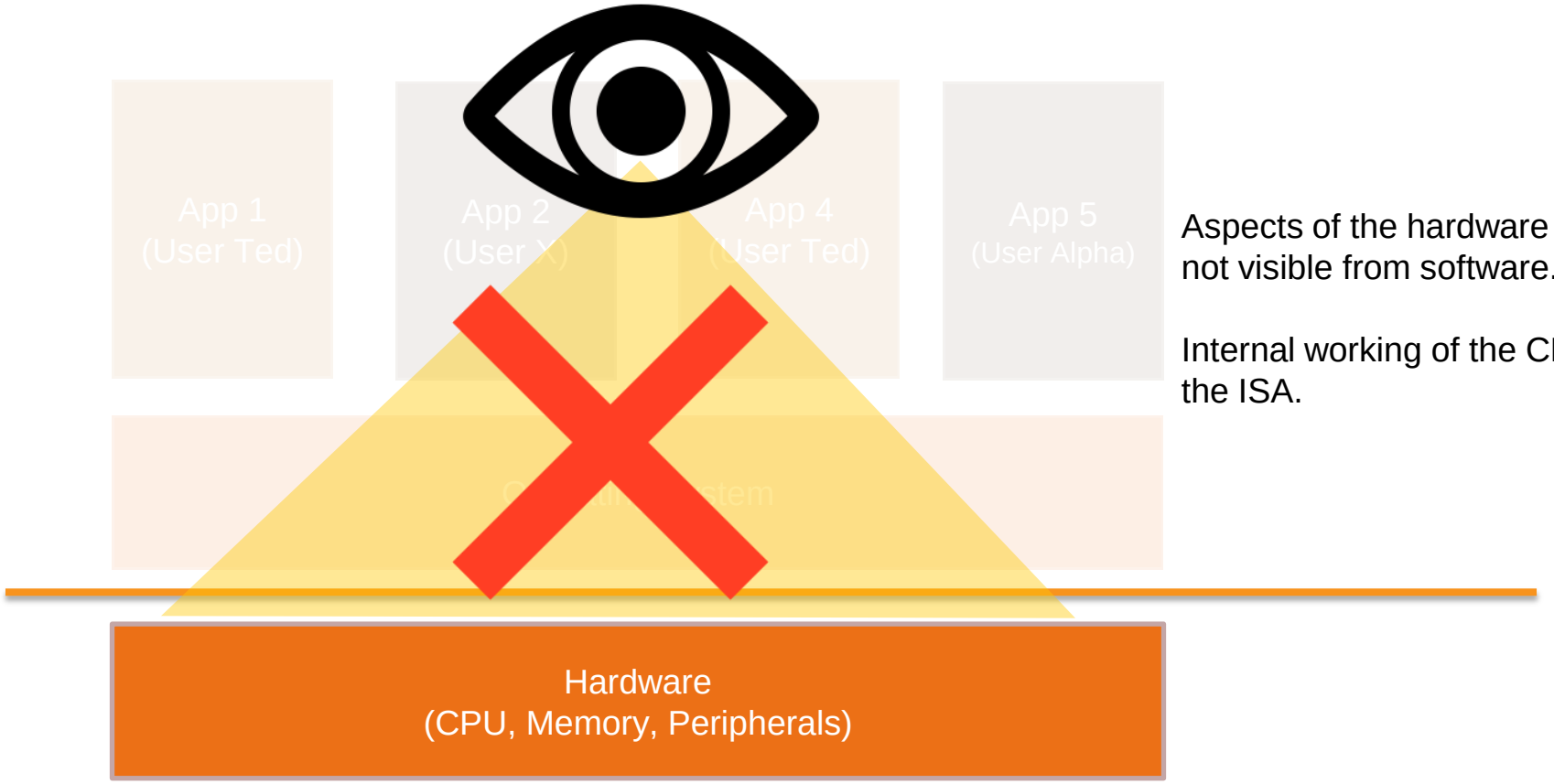


Aspects of the hardware visible from software.

An abstraction of the hardware

Instruction Set Architecture (ISA) + Operating System Interface

Computer Organization



Aspects of the hardware not visible from software.

Internal working of the CPU that realizes the ISA.

Two processors can have the same ISA but different implementations of the architecture, with varying cost and performance.

Microprocessors

General Purpose



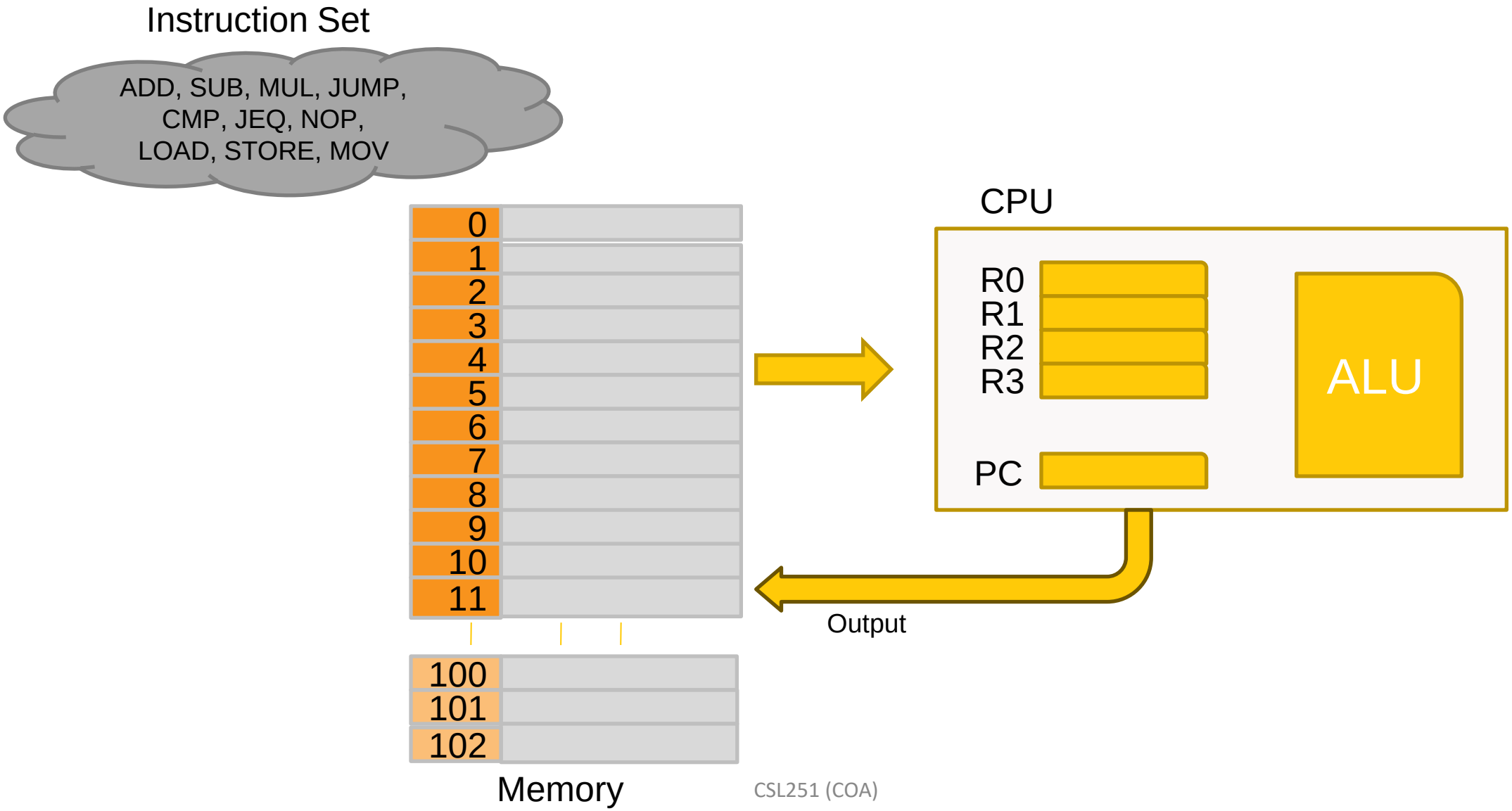
Application Specific



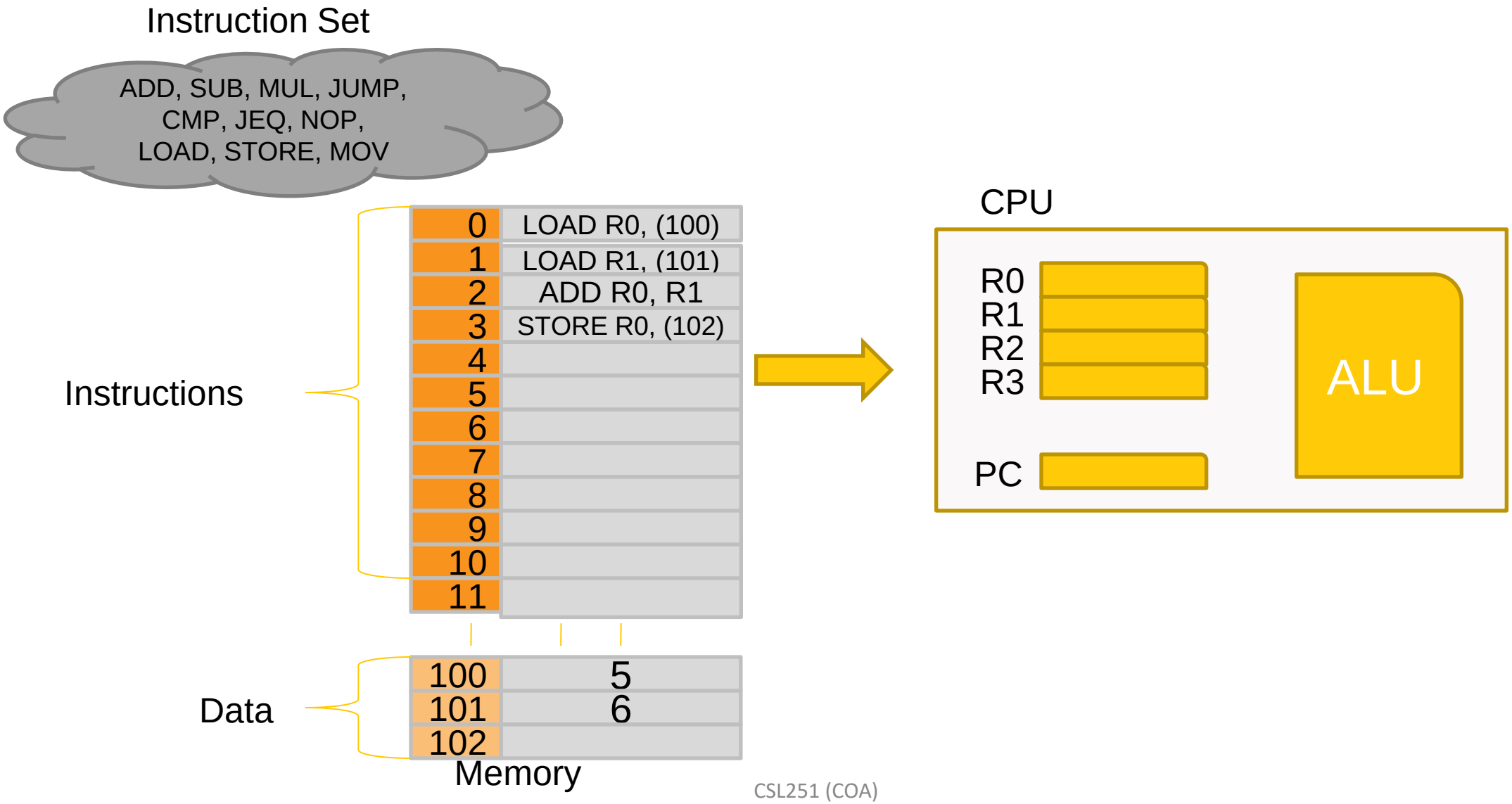


Many different processors, but underlying concept used is the same: Stored Program Concept

Stored Program Concept



Stored Program Concept



This Course (theory and Lab)

- How does the processor support programming languages?
- How does the processor support operating systems?
- How to design a processor to improve speed and performance?
- How to design a processor to support multiple users with limited resources?
- Lab: Assembly coding, processor design / analysis



Logistics and Grading

Instructor	**Dr. Sk Subidh Ali(subidh@iitbhilai.ac.in)**		
Credit	4	**Prerequisite(s)**	CSL100 and ECL 101
Classes	Tuesday	9:30-10:25 AM	
	Wednesday	10:30-11:25 AM	
	Friday	10:30-11:25 AM	
Lab	Monday	03:30-05:25 PM	

Reference Books

